

TS 2018 Class 12 Physics**Paper-II****Time allowed: 3 hours****Maximum Marks : 60****SECTION-A $(10 \times 2 = 20)$** **Note: i) Answer all the questions.****ii) Each question carries two marks.****iii) All are very short answer type questions.****1. What is hypermetropia ? How can it be corrected?**

Ans: Hypermetropia (long-sightedness) is common eye conditions that are occur when light does not focus on the retina of the eye. This can be easily corrected using prescription glasses or contact lenses or in mild cases by laser eye surgery.

2. How do you convert a moving coil galvanometer into an ammeter ?

Ans: A galvanometer can be converted into an ammeter by connecting a low resistance (called shunt resistance) in parallel to the galvanometer.

3. What is the magnetic moment associated with a solenoid?

Ans: Magnetic Field of a Solenoid. You can create a stronger, more concentrated magnetic field by taking wire and forming it into a coil called a solenoid. ... In the presence of a magnet, these filings will rearrange themselves according to the magnetic lines of force (flux lines) so are very commonly used as switches.

4. Define magnetic declination.

Ans: It is defined as the angle on the horizontal plane between magnetic north and true north. The angle varies depending on position on the Earth's surface and changes over time.

5. Write the expression for resistance of an inductor and a capacitor.

Ans: By using Ohm's law, which states that resistance equals voltage divided by current, or

$R = V/I$ (more commonly written as $V = IR$), where R is resistance, V is voltage and I is current. Resistors are generally classified as either fixed or variable.

6. What are the applications of microwaves?

Ans: Microwaves are most commonly used in

1. satellite communications,
2. radar signals, phones, and
3. navigational applications.
4. medical treatments,

5. drying materials, and in
6. households for the preparation of food.

7. Write De Broglie's relation and explain the terms therein.

Ans: The de Broglie equation is an equation used to describe the wave properties of matter, specifically, the wave nature of the electron:

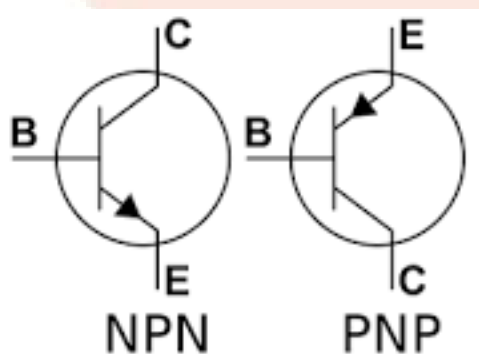
- $\lambda = h/mv$, where λ is wavelength, h is Planck's constant, m is the mass of a particle,
- moving at a velocity v

8. What is work function.

Ans: Minimum thermodynamics work which is used to eliminate an electron from solid to a point in vacuum immediately outside the surface.

9. Draw the circuit symbols for p-n-p, and n-p-n transistors.

Ans:



10. Mention the basics methods of modulation.

Ans: Modulation refers to the variation in strength, tone of

individual's voice. It generally categorized into four basic types. These modulations are -

1. Analog modulation
2. Digital modulation
3. Pulse modulation
4. Spread spectrum method.

SECTION- B

(6 × 4 = 24)

Note: i) Answer Any Six of the following questions.

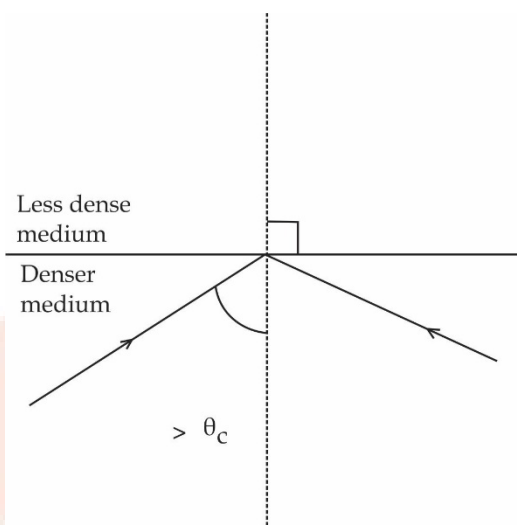
ii) Each question carries four marks.

iii) All are short answer type questions

11. Define critical angle . Explain total internal reflection using a neat diagram.

Ans: The angle of incidence for which angle of refraction is 90° is known as critical angle.

When a ray of light in medium is reflected back into the medium is known as total internal reflection. Generally, this phenomenon is occurred when the angle of incidence is greater than the critical angle.



12. How do you determine the resolving power of your eye.

Ans: An instrument which is designed to see the reflexes of an intra-molecular line known as retinoscopy. Lens reflex is observed when the light is flashed on the pupil of an eye.

13. State Gauss's law in electrostatics and explain its importance.

Ans: The total electric flux out of closed surface is equal to charge enclosed divide by permittivity is known as the Gauss's law.

Importance of gauss's law are: It is used to find the electric field which is enclosed in a closed surface.

14. Derive an expression for the capacitance of a parallel plate capacitor.

Ans: Capacitance of a parallel plate capacitor:

$$\sigma = \frac{Q}{A}$$

The electric field strength between the plates

$$E = \frac{\sigma}{K\epsilon_0}, \text{ where } \epsilon_0 = \text{permittivity of free space ... (i)}$$

The potential difference between the plates,

$$V_{AB} = Ed = \frac{\sigma d}{K\epsilon_0} \dots\dots\dots \text{(ii)}$$

Putting the value of σ we get

$$V_{AB} = \frac{(Q/A)d}{K\epsilon_0} = \frac{Qd}{K\epsilon_0 A}$$

$\therefore$ Capacitance of capacitor

$$C = \frac{Q}{V_{AB}} = \frac{Q}{(Qd / K\epsilon_0 A)}$$

$$C = \frac{K\epsilon_0 A}{d}$$

15. State and explain amperes law .

Ans: Ampere's law states that the magnetic field which is formed by the electric current is proportional to the size of that electric current with a constant of proportionality.

Permeability of
free space (1.26×10^{-6})

Magnetic field
in teslas, T

Current in the
wire, in amps, A

$$B = \frac{\mu_0 I}{2\pi r}$$

Radius from the
wire, in meters, m

16. Describe the ways in which eddy current are used to advantage .

Ans: Eddy currents are used in many ways:

1. the brakes of eddy currents are used to stop or slow the moving object.
2. Eddy currents are also used as a crack detection equipment. It can be divided into high frequency and the low frequency. In high frequency equipment it is used to detect the crack in shining or non-shining materials.

In the low frequency equipment, the process known as the induction furnace is used to separate the material from the ores.

17. Describe Rutherford atomic model . What are the drawbacks of this model ?

Ans: According to the Rutherford atomic model, most of the space in atom is not filled. Charges of the atom is concentrated in small circular structure known as nucleus. This nucleus is

surrounded by the electrons (negatively charged particles).

Drawbacks of Rutherford atomic model:

- Rutherford atomic model does not explain the stability of atoms.
- It also enables to explain that how moving electrons remain in its orbit.

18. Distinguish between half and full wave rectifier.

Ans: Half wave rectifier conducts current only during positive half cycle of the input whereas full wave rectifier conducts both positive as well as negative half cycle of the input.

1. Output frequency of half wave rectifier is equal to the frequency of input whereas in full wave rectifier output frequency is twice of the input.
2. Only one diode is required for half wave rectifier whereas two diodes are required for full wave rectifier.
3. For half wave rectifier, there is no need for center tapped transformer whereas center tapped transformer is required for full wave rectifier.
4. Efficiency of half wave rectifier is 40.6% whereas efficiency of full wave rectifier is 81.2%

SECTION –C **(2 × 18 =16)**

Note: i) Answer ANY TWO of the following questions.

ii) Each question carries Eight marks.

iii) All are long answer type questions

19. What is doppler effect ? when the source is in motion with respect to an observer at rest . mention its applications.

Ans: The changes in the frequency and wavelength, which is caused by the change in distance between the sender and observer is known as Doppler's effects. When the source is in motion with respect to an observer at the rest, then the waves are closer to the observer which leads to the causing an increase in the frequency.

Applications of Doppler's effects:

- Doppler's effect is used in the astronomy, where it used to measure the speed at which stars are moving.
- It is also used in the medical imaging for determine the direction of blood flow in arteries and veins.
- This effect is used in the radar to measure the velocity detected objects where radar beam is fired at target.

20. State Kirchhoff's laws for an electrical network. Using these laws deduce the condition for the balance in a Wheatstone Bridge.

Ans: Kirchhoff's first law (junction rule) - In an electrical circuit, the algebraic sum of currents meeting at a junction, is always zero.

Kirchhoff's second rule (loop rule) - In any closed mesh of an electrical circuit, the algebraic sum of the e.m.f is equal to the algebraic sum of products of resistances and current flowing through them.

Wheatstone bridge (balanced) - Let i be the current from battery E. At point A, current i flows through resistance p and current $i - i_1$ flows through R . In balanced state, no current flows through BD , hence point B and D are at same potential. Therefore current i_1 flows through resistance Q also and current $i - i_1$ flows through S .

Applying Kirchhoff's loop rule in closed mesh ABDA.

$$i_1 P = (i - i_1) R \dots\dots\dots \text{eq1}$$

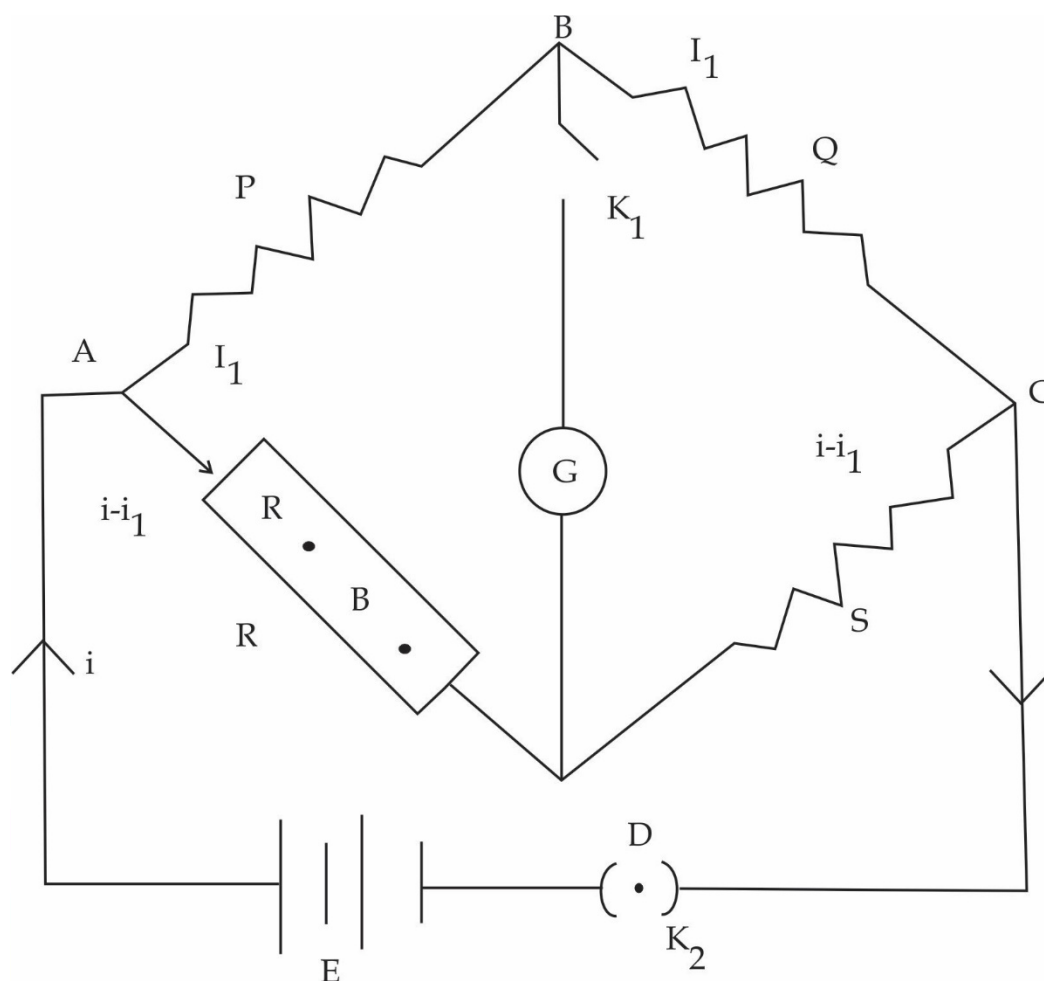
In closed mesh BCDB

$$i_1 Q = (i - i_1) S \dots\dots\dots \text{eq2}$$

Dividing eq1 from eq2, we get

$$P/Q = R/S$$

This is the condition for balance in a Wheatstone's bridge



21. Explain the principle and working of a nuclear reactor with the help of a labelled diagram.

Ans. Nuclear reactor:

Principle: It works on the principle of achieving controlled chain reaction of Uranium ^{238}U enriched with ^{235}U and as a result generating huge amount of energy. The controlled chain is made possible by:

slowing down the fission neutrons to thermal neutrons by using a moderator.

inserting rods that can absorb neutrons (control rods) and use them to maintain the reaction rate.

Working: -

Clearly explained in the image.

In general the fuel used is ^{235}U which is filled in aluminium cylindrical rod.

The moderator used are D_2O heavy water, salt water and graphite block.

The materials like Beryllium and Cadmium rods are used as control rods which controls the chain reaction by absorbing the neutron emitted in the nuclear reaction.

Cold water is circulated through the water pipes surrounding the nuclear reactor to absorb heat.

